

Vol 3 Chapter 7 — Atomic Orbital Sequence:

$$(2l + 1) = 1, N_c, n_C, g$$

Keeper (author pass)

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Chapter 7 — Atomic Orbital Sequence

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

The atomic-orbital structure of multielectron atoms follows the famous sequence $1s, 2s, 2p, 3s, 3p, 3d, 4s, \dots$ — with orbital occupancies of $(2l + 1) = 1, 3, 5, 7$ electron-pairs per shell type. The pattern is empirical in standard atomic physics; it emerges from the hydrogenic shell structure modified by electron-electron interactions.

BST identifies the orbital sequence with BST primary integers exactly:

$$(2l + 1) = 1, N_c, n_C, g = 1, 3, 5, 7.$$

D-tier exact. The atomic-orbital sequence is the substrate's operator-zoo sequence (Volume 0 Chapter 7) applied to the atomic scale.

7.1 The structural identification

The four atomic orbital types (s, p, d, f) correspond to angular-momentum quantum numbers $l = 0, 1, 2, 3$. The orbital occupancies $(2l + 1)$ give $1, 3, 5, 7$ — exactly the sequence $1, N_c, n_C, g$ of BST primaries (with the trivial 1 corresponding to the s -orbital scalar K-type).

This identification is structurally significant. The atomic shell structure is not separate physics — it is the substrate's K-type angular-momentum spectrum applied at the atomic energy scale.

7.2 Consequence: the periodic table

The periodic table's structure — eight-column blocks for $s + p$ (8 electrons), eighteen-column blocks for $s + p + d$ (18 electrons), thirty-two-column blocks for $s + p + d + f$ (32 electrons) — emerges directly from the substrate's 1, 3, 5, 7 K-type degeneracies. The periodic table is the substrate's orbital-spectrum at the atomic scale.

7.3 Match precision

D-tier exact. The atomic-orbital sequence matches integer-exact to the BST primary sequence.

7.4 What comes next

Chapter 8 — Hyperfine + Lamb Shift — develops the $\alpha^{\text{BST primary}}$ exponent pattern in atomic spectroscopy.

Where to look this up: Atomic orbital substrate-anchor: Vol 0 Ch 7 + Vol 1 Ch 6.
For standard atomic physics: Bransden and Joachain.